

MUDANJIANG, China

WMO: 540940

Lat: 44.5039N

Lon: 129.6683E

Elev: 1006

StdP: 14.17

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.2	-11.3	-22.3	1.7	-12.3	-19.1	2.0	-8.8	23.0	10.2	20.4	9.9	2.0	200	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	88.6	70.4	86.1	69.2	83.4	68.5	74.7	83.5	73.1	81.0	71.6	79.2	7.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.0	122.7	79.7	70.5	116.7	78.1	69.0	110.8	76.7	39.0	83.7	37.5	81.0	36.2	79.2	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	17.8	14.8	DB	-18.6	93.8	5.6	2.5	-22.6	95.6	-25.8	97.0	-28.9	98.4	-33.0	100.2	(4)
(5)			WB	-18.9	77.1	5.5	1.6	-22.9	78.3	-26.1	79.2	-29.1	80.1	-33.1	81.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.4	3.4	12.2	27.4	44.9	58.0	68.4	73.4	71.0	59.8	44.3	24.7	8.1		
	DBStd	25.62	7.39	9.27	9.68	8.54	7.32	6.15	4.59	4.97	6.25	8.06	10.88	8.31		
	HDD50	5697	1445	1058	701	201	14	0	0	0	5	214	760	1299		
	HDD65	9236	1910	1478	1165	603	240	35	3	10	176	643	1209	1764		
	CDD50	2573	0	0	1	49	261	551	724	650	299	37	1	0		
	CDD65	637	0	0	0	1	22	136	262	195	21	0	0	0		
	CDH74	4953	0	0	0	33	344	1186	1950	1270	165	5	0	0		
	CDH80	1453	0	0	0	5	87	392	638	315	16	0	0	0		
	WSAvg	5.2	4.7	5.2	6.4	6.6	6.1	5.0	4.4	4.2	4.3	5.1	5.3	4.8		
	PrecAvg	21.1	0.2	0.2	0.4	1.1	1.9	3.5	4.6	4.8	2.2	1.3	0.6	0.3		
(15) Precipitation	PrecMax	29.4	0.6	0.7	1.3	3.0	5.3	8.0	11.6	13.7	4.4	3.9	2.1	1.2		
	PrecMin	13.3	0.0	0.0	0.0	0.1	0.2	0.5	0.8	1.4	0.3	0.1	0.0	0.0		
	PrecStd	4.1	0.1	0.2	0.3	0.7	1.1	1.8	2.2	2.9	1.2	0.9	0.4	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.1	42.0	59.4	78.3	87.1	92.4	92.9	89.6	82.2	72.8	57.6	39.1		
		MCWB	24.5	33.2	44.2	55.3	60.8	66.8	73.2	74.4	67.2	57.2	46.1	34.1		
	2%	DB	22.6	37.4	52.0	71.9	82.3	87.9	89.0	86.4	78.7	67.0	50.6	29.9		
		MCWB	18.8	30.6	39.8	51.8	59.2	66.5	72.1	72.4	64.3	53.3	41.7	25.7		
	5%	DB	18.6	32.7	47.0	66.8	77.8	84.7	86.3	83.5	75.2	62.3	45.2	25.0		
		MCWB	15.6	27.3	36.9	49.3	57.9	65.8	71.4	70.9	62.1	50.9	38.1	21.7		
	10%	DB	15.0	28.2	42.4	61.5	73.0	80.9	83.6	80.8	72.1	57.9	40.2	20.8		
		MCWB	12.5	23.5	34.3	46.2	55.8	64.9	70.8	69.7	60.7	48.5	34.5	17.9		
	0.4%	WB	24.8	34.6	45.3	58.0	63.8	71.9	77.6	76.6	70.1	58.5	47.2	34.2		
		MCDB	29.2	40.4	57.0	74.1	81.9	83.2	87.6	86.7	78.5	69.5	55.9	38.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	19.2	30.9	40.9	52.8	61.1	69.9	75.3	74.6	67.2	55.2	42.6	26.0		
		MCDB	22.5	36.8	50.1	69.6	77.1	82.3	84.7	83.0	74.8	63.5	49.5	29.5		
	5%	WB	15.6	27.3	37.7	50.0	59.1	68.2	73.7	73.1	64.6	52.5	38.7	21.8		
		MCDB	18.4	32.6	45.7	65.5	73.5	79.7	82.3	80.3	72.4	60.6	44.8	24.9		
	10%	WB	12.5	23.5	34.7	47.2	57.2	66.5	72.2	71.6	62.4	49.5	34.9	18.0		
		MCDB	14.9	28.1	42.0	60.0	70.7	77.9	80.4	78.1	70.1	56.6	40.1	20.8		
	5% DB	MDBR	18.7	21.2	19.1	21.4	21.1	19.4	16.6	16.7	21.6	21.0	17.3	16.7		
		MCDBR	21.8	23.9	24.6	30.2	30.3	25.3	21.9	20.5	25.1	27.6	23.2	19.9		
		MCWBR	18.6	18.2	14.9	15.1	12.0	9.0	8.4	8.4	11.8	14.9	15.7	16.7		
		MCWBR	21.9	23.0	22.8	28.2	25.9	21.1	17.6	16.9	20.3	24.5	22.5	19.5		
(35) Clear-Sky Solar Irradiance	taub		0.326	0.366	0.452	0.547	0.583	0.572	0.540	0.492	0.440	0.463	0.388	0.340		
	taud		2.051	1.951	1.827	1.719	1.705	1.818	1.948	2.066	2.116	1.923	2.016	2.027		
	Ebn at Noon		238	248	241	229	226	230	236	243	245	212	209	215		
	Edh at Noon		33	44	58	70	73	66	57	49	43	45	35	31		
(44) All-Sky Solar Radiation	RadAvg		706	1061	1373	1480	1635	1742	1586	1491	1351	957	645	567		
	RadStd		59	53	85	114	145	186	144	130	107	73	49	35		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page